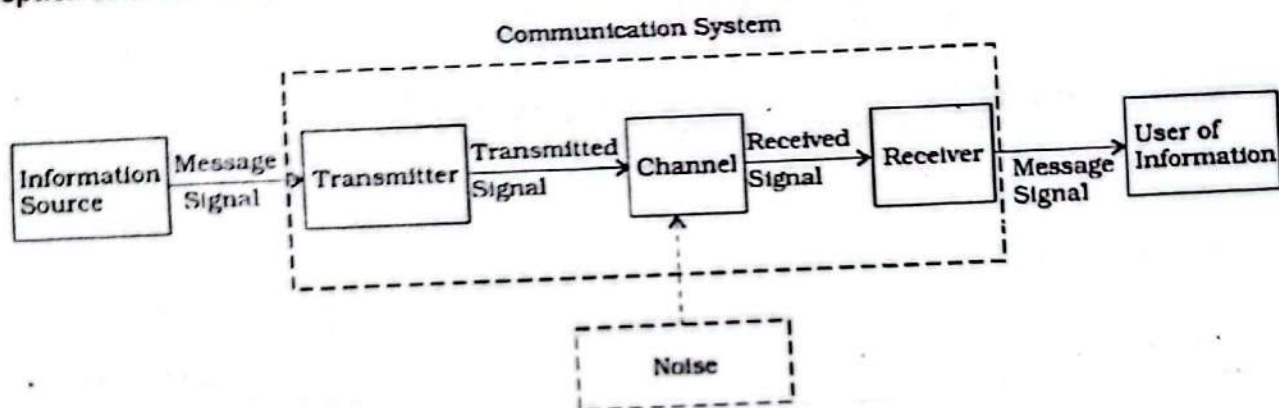


CHAPTER 14

Wired and Wireless Communication

COMMUNICATION: is the process of transmitting information or signals from one point to another using various method or channels. The steps involve the transmission, channel or medium and reception of the message.



BLOCK DIAGRAM OF COMMUNICATION SYSTEM

Transmitter:

A transmitter is a device or system in communication that converts information or signals into a form which is suitable for transmission over a communication channel. It typically generates and amplifies electromagnetic signals.

Receiver

A receiver is a device or system in communication that captures and processes signals transmitted over a communication channel. It is responsible for detecting and demodulating the received signals to recover the original information or data.

Channel:

Channel refers to the medium or pathway through which information or data is transmitted from a sender to a receiver. It is the physical or virtual means by which signals, messages, or information are conveyed. Channels can vary widely in terms of their characteristics and suitability for different types of communication. They are divided into Wired and Wireless communications channels.

Wired Channel: Wired communication channels use physical wires or cables to transmit data. Examples include Ethernet cables and telephone lines.

The component used for wired channels now a days are:

- Twisted Pair Cable
- Coaxial Cable
- Optical Fiber Cable

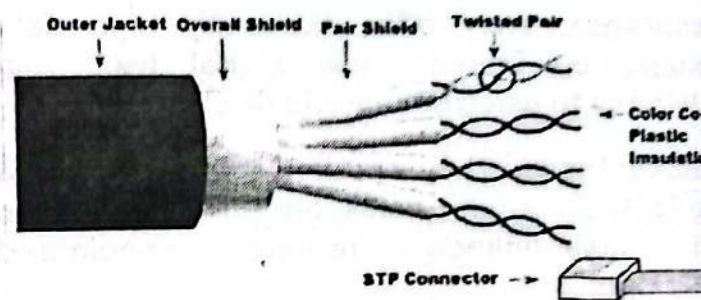
Twisted Pair Cable:

Description: Twisted pair cable consists of pairs of insulated copper wires twisted together. It is one of the most common types of communication cables used for various applications, including telephone lines and Ethernet networking.

Use: Twisted pair cables are suitable for short to medium-range data transmission and are commonly used in both residential and commercial settings.

Advantages: They are cost-effective, easy to install, and resistant to electromagnetic interference.

Disadvantages: They have limited bandwidth compared to other cable types, making them less suitable for high-speed data transmission over long distances.



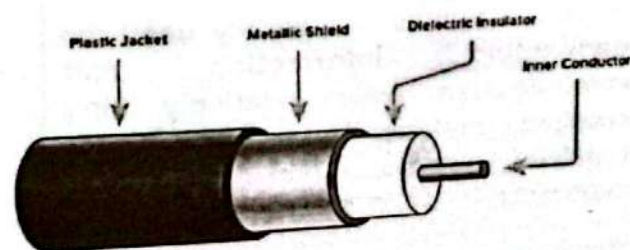
Coaxial Cable:

Description: Coaxial cable consists of a central conductor surrounded by a dielectric insulator, a metallic shield, and an outer insulating layer. It is commonly used for cable television (CATV) and broadband internet.

Use: Coaxial cables provide higher bandwidth and are suitable for transmitting television signals, internet data, and other high-frequency signals.

Advantages: They offer better shielding and can transmit signals over longer distances without significant loss.

Disadvantages: They are thicker and less flexible than twisted pair cables, which can make installation more challenging.



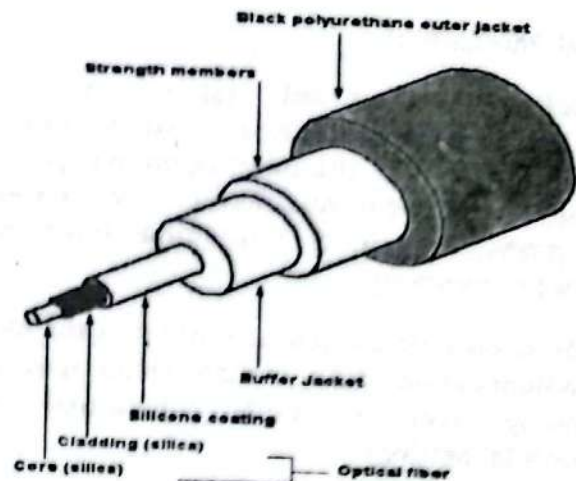
Optical Fiber Cable or Fiber Optic Cable

Description: Optical fiber cable uses thin strands of glass or plastic fibers to transmit data using light signals. It is known for its high bandwidth and data transmission capabilities.

Use: Optical fiber cables are widely used in telecommunications, high-speed internet, and long-distance data transmission.

Advantages: They offer extremely high data transmission speeds, low signal loss, and resistance to electromagnetic interference.

Disadvantages: They are more expensive and fragile than copper cables, and their installation and maintenance require specialized equipment.



Wireless Channel: Wireless communication channels transmit data without physical wires or cables. Examples include Wi-Fi, Bluetooth, and cellular networks.

The EM Waves used for wireless channels now a days are :

- Radio Waves
- Microwaves

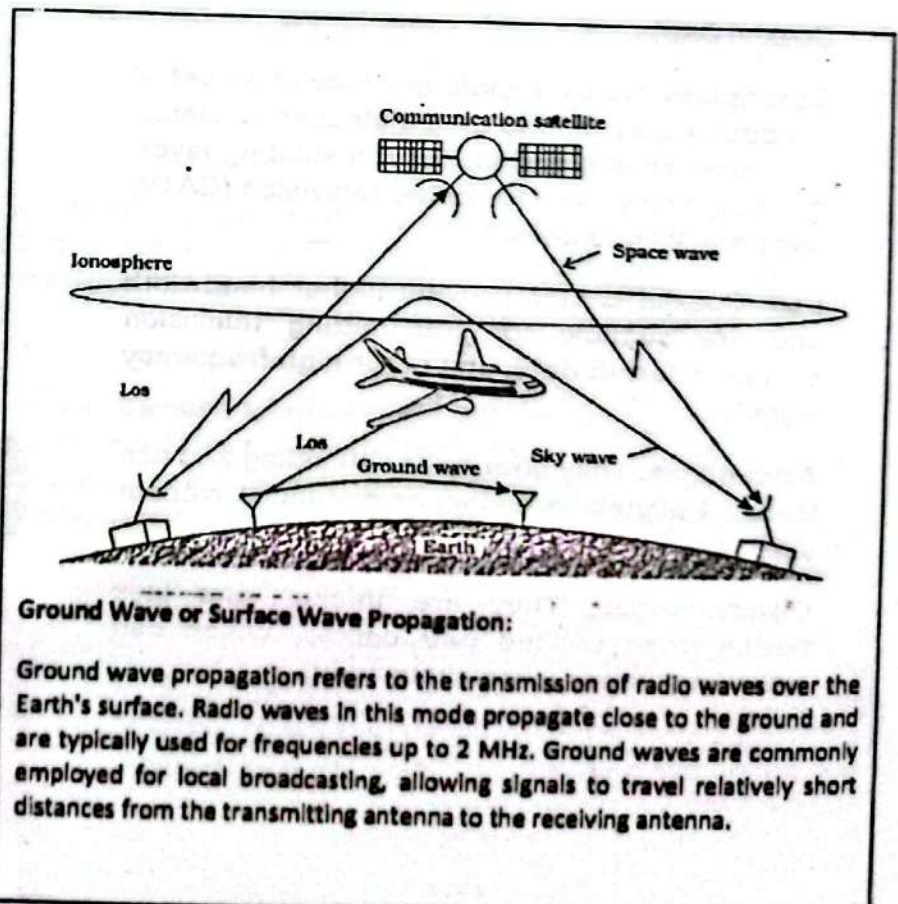
Radio Waves:

Description: Radio waves are a type of electromagnetic wave with a long wavelength. They are used for wireless communication and broadcasting, including AM and FM radio, TV broadcasting, and cellular networks.

Use: Radio waves are widely used for broadcasting information and communication over relatively long distances, making them suitable for terrestrial radio and television broadcasts.

Advantages: They have good propagation characteristics, allowing them to travel through obstacles like buildings and trees, making them suitable for wide-area coverage.

Disadvantages: They have limited bandwidth, which can affect data transmission speeds.



Microwaves:

Description: Microwave communication is a method of transmitting data and signals using electromagnetic waves with relatively short wavelengths, typically in the microwave frequency range (wavelengths between approximately 1 millimeter and 1 meter). These waves are part of the electromagnetic spectrum, with frequencies ranging from 1 GHz (gigahertz) to 300 GHz.

Use: Microwave communication is commonly used for various applications, including point-to-point and point-to-multipoint communication links. It is often used in terrestrial communication systems, such as microwave towers and parabolic dish antennas, to transmit data, voice, and video signals over short to medium distances.

Advantages: Microwaves provide high bandwidth and enable long-distance communication between ground stations and satellites.

Disadvantages: They are affected by atmospheric conditions, such as rain, which can attenuate the signal.

Sky Wave Propagation:

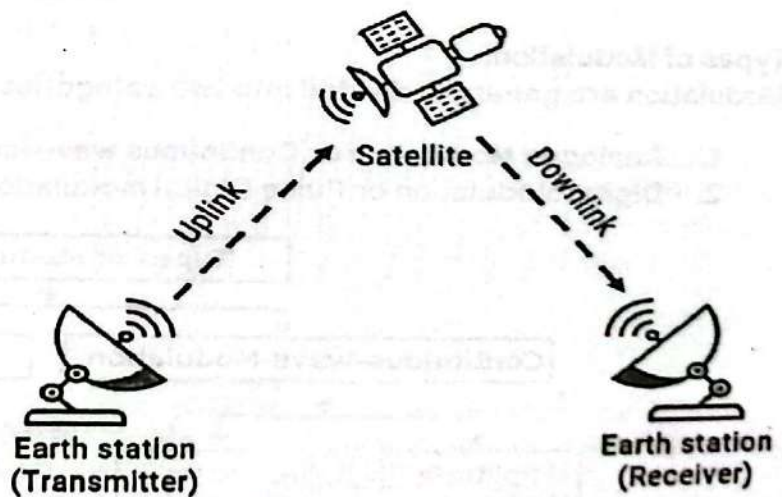
Sky wave propagation, also known as ionosphere propagation, involves the reflection or refraction of radio waves by the Earth's ionosphere. This mode is suitable for frequencies ranging from 3 MHz to 30 MHz and is primarily used for long-distance radio communication. By bouncing off the ionosphere, these waves can cover extensive distances.

Space Wave Propagation:

Space wave propagation occurs in the frequency range of 30 MHz to 300 MHz. In this mode, radio waves travel directly from the transmitting antenna to the receiving antenna without significant reflection or refraction. It is often referred to as line-of-sight communication because the waves travel in a straight line. Space wave propagation is commonly utilized in television communication and radar systems.

Satellite Communication:

Satellite communication involves the use of artificial satellites orbiting the Earth to relay signals and data between ground stations or other satellites. Communication satellites are equipped with transponders that receive, amplify, and retransmit signals to provide global coverage. Satellite communication is used for a wide range of applications, including long-distance data transmission, television broadcasting, internet connectivity, and global positioning systems (GPS). It enables communication over vast distances and remote areas where terrestrial infrastructure is limited.



CHAPTER 14

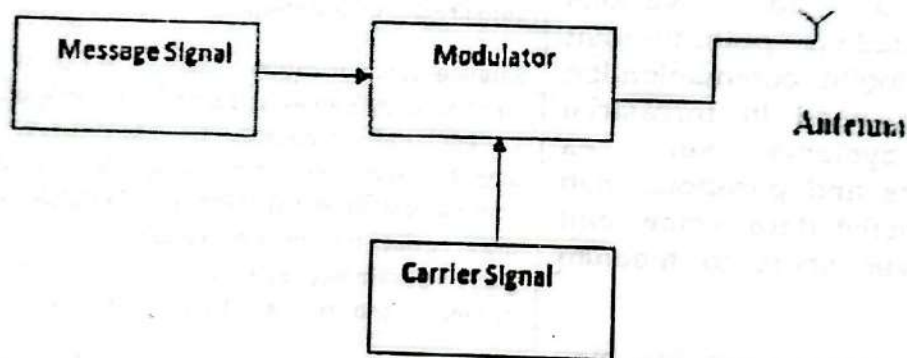
Modulation (Amplitude Modulation)

Modulation:

Modulation is a process by which some of the characteristics such as amplitude, frequency or wavelength of the Carrier signal are varied in accordance with the modulating signal (message signal). The purpose of modulation is to transmit message over a long distance and to transmit multiple message signal over a common channel.

Carrier Signal: A Carrier Signal is a continuous, high-frequency signal that is used as a carrier wave to transmit information or data

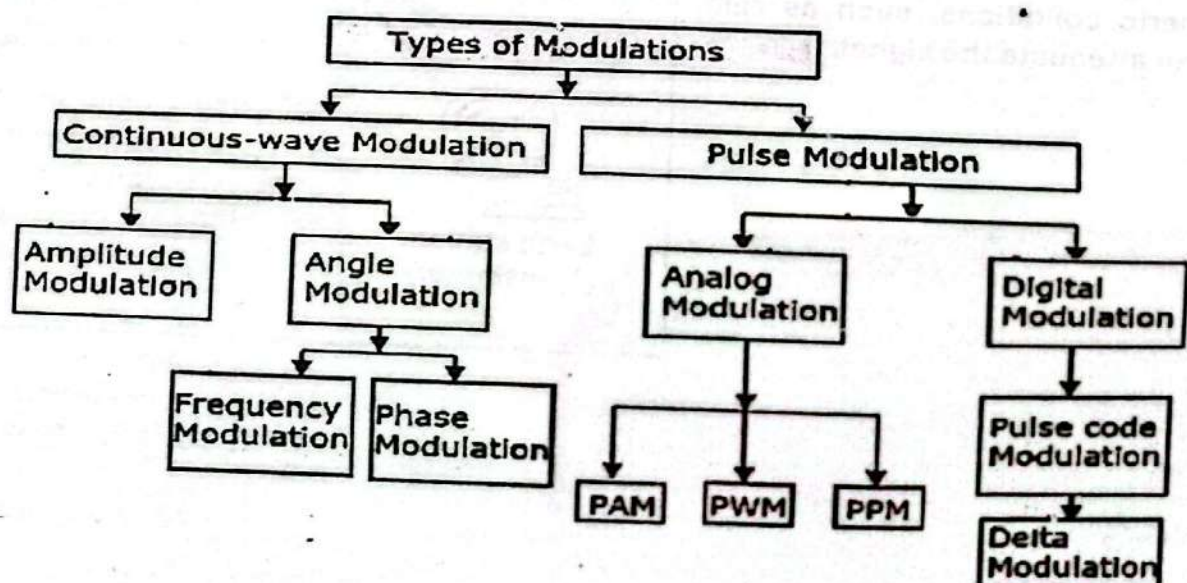
Message Signal: A Message Signal also known as the modulating signal or baseband signal, is the signal that carries the actual information or data that needs to be transmitted



Types of Modulation:

Modulation are generally divided into two categories

1. Analogue Modulation or Continuous wave modulation
2. Digital Modulation or Pulse Digital modulation



AMPLITUDE MODULATION:

Amplitude Modulation (AM) is a modulation technique in which the carrier signal amplitude is varied with respect to the amplitude of the message signal

Derivation:

Consider a message signal (baseband signal) $m(t)$ and a carrier signal $c(t)$ which need to be amplitude modulated resulting a modulated wave $AM(t)$

Message signal:

$$m(t) = A_m \cos(2\pi f_m t)$$

Carrier signal:

$$c(t) = A_c \cos(2\pi f_c t)$$

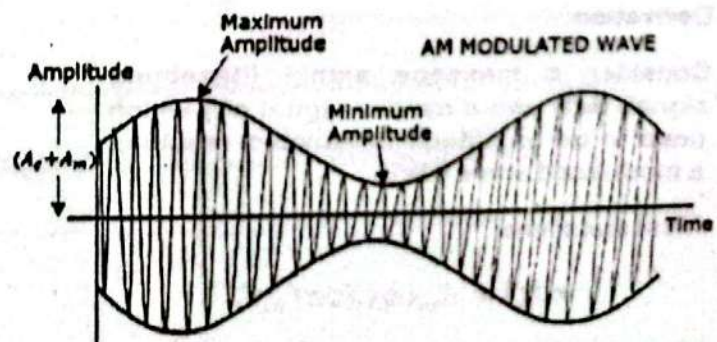
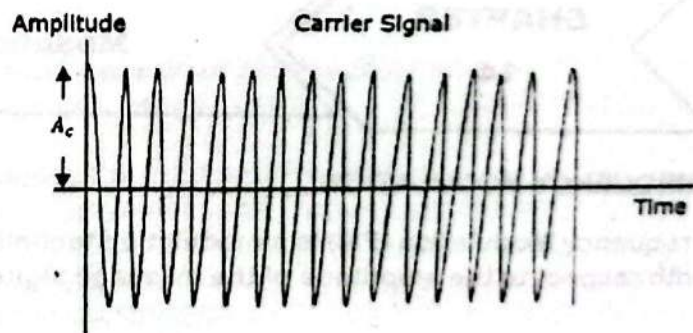
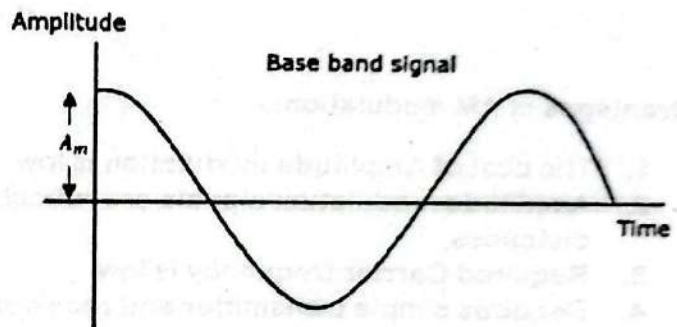
Where,

A_m and A_c are the amplitude of the modulating signal and the carrier signal respectively.

f_m and f_c are the frequency of the modulating signal and the carrier signal respectively.

As we know in amplitude modulation the amplitude of the carrier signal is changed. Therefore

$$AM(t) = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$$



Modulation Index:

A carrier wave, after being modulated, if the modulated level is calculated, then such an attempt is called as Modulation Index or Modulation Depth. It states the level of modulation that a carrier wave undergoes

Rearranging above Eq. we get;

$$AM(t) = A_c \left[1 + \left(\frac{A_m}{A_c} \right) \cos(2\pi f_m t) \right] \cos(2\pi f_c t)$$

$$AM(t) = A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

Where, μ is Modulation Index and it is equal to the ratio of A_m and A_c . Mathematically, we can write it as

$$\mu = \frac{A_m}{A_c}$$

Advantages of AM Modulation:

1. The cost of Amplitude modulation is low
2. Amplitude modulation signals are reflected by the earth ionosphere thus can travel long distances.
3. Required Carrier frequency is low
4. Requires simple transmitter and receiver.

CHAPTER 14

Modulation (Frequency Modulation)

FREQUENCY MODULATION:

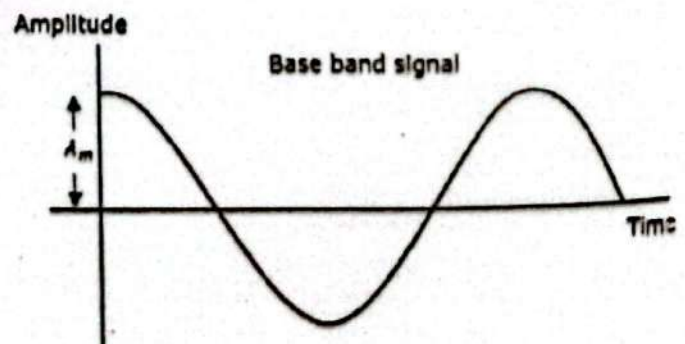
Frequency Modulation (FM) is a modulation technique in which the carrier signal frequency is varied with respect to the amplitude of the message signal

Derivation:

Consider a message signal (baseband signal) $m(t)$ and a carrier signal $c(t)$ which need to be amplitude modulated resulting a modulated wave $FM(t)$

Message signal:

$$m(t) = A_m \cos(2\pi f_m t)$$

Carrier signal:

$$c(t) = A_c \cos(2\pi f_c t)$$

Where,

A_m and A_c are the amplitude of the modulating signal and the carrier signal respectively.

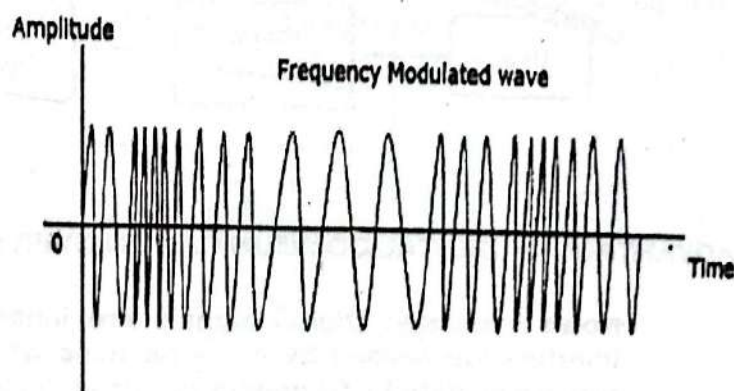
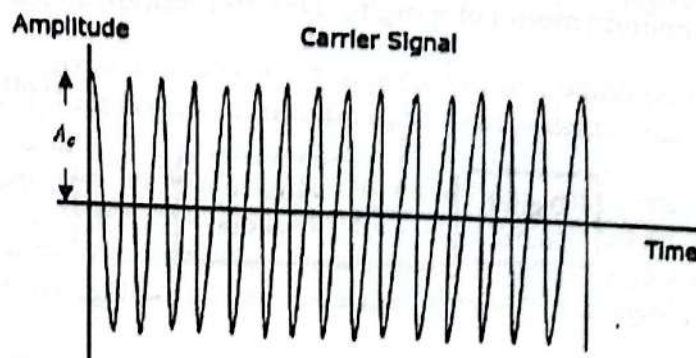
f_m and f_c are the frequency of the modulating signal and the carrier signal respectively.

As we know in frequency modulation the frequency of the carrier signal is changed. Therefore

$$FM(t) = f_c + kA_m \cos(2\pi f_m t)$$

$$FM(t) = f_c + km(t)$$

Where, $FM(t)$ is frequency modulated wave
 f_c is frequency of carrier signal
 $m(t)$ is modulating signal
 k is proportionality constant



Advantages of AM Modulation:

1. The amplitude of the frequency-modulated wave does not get affected.
2. Frequency Modulation decreases the noise hence, there is a significant increase in the signal to noise ratio.
3. We can also reduce the noise by increasing the frequency deviation.
4. It also reduces the interference by the adjacent channels through guard bands.
5. It operates in a very high frequency called VHF.
6. It is resistant to signal strength variations.
7. It does not require a linear amplifier in the transmitter.

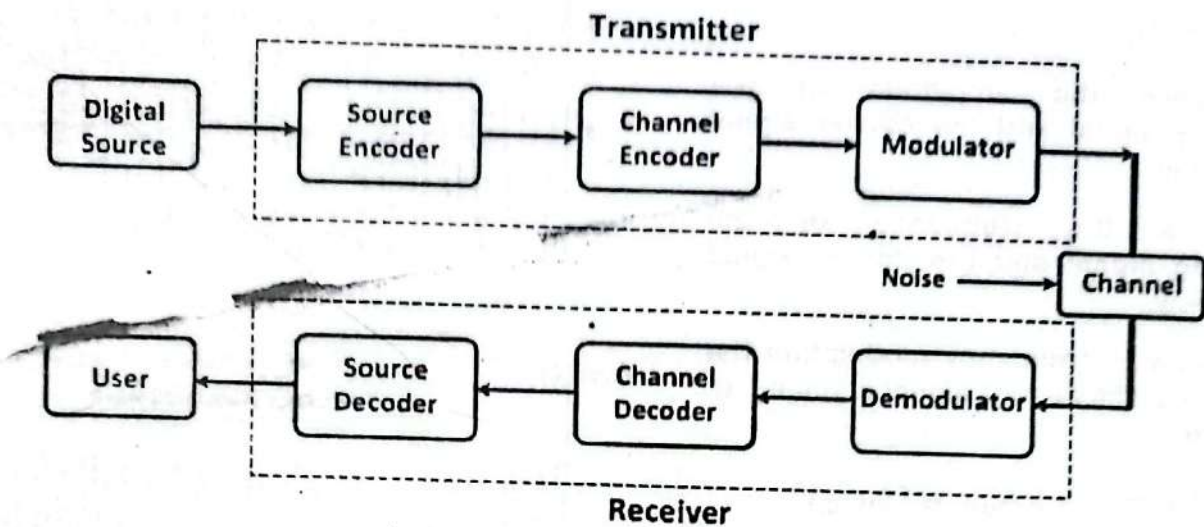
CHAPTER 14

Digital Communication Systems

DIGITAL COMMUNICATION SYSTEMS

A digital communication system is a method of transmitting information or data in the form of discrete, binary symbols (0s and 1s) over a communication channel. It's a modern and efficient way of transmitting data and has largely replaced analog communication systems in many applications.

The general model of a digital communication system:



ADVANTAGE OF DIGITAL COMMUNICATION OVER ANALOG COMMUNICATION:

- **Noise Immunity:** Digital signals are inherently less susceptible than analog signals to interference caused by noise because with digital signals it is not necessary to evaluate precise amplitude, frequency or phase. Instead pulses are evaluated during the precise time interval and simple determination is made whether the pulse is above or below a prescribed reference level.
- **Multiplexing:** Digital signals are better suited than analog signals for processing and combining using a technique multiplexing.
- **Easy to Store:** It is simple to store digital signals than analog signals.
- **Resistant to additive Noise:** Digital transmission systems are more resistant to analog system to additive noise because they use signal regeneration rather than signal amplification. Noise produced in electronic circuit is additive, therefore S/N ratio deteriorates each time an analog signal is amplified.
- **Used for Long Distance:** Digital regenerators sample noisy signals and then reproduce an entirely new digital signal with same S/N ratio as the original transmitted signals. So digital transmitted signals can be transported longer distance than analog signals.
- **Transmission errors can be detected easily:** The transmission errors can be detected and corrected more easily and accurately than is possible with analog signals.

DIGITAL TRANSMISSION OF ANALOG SIGNAL:

In order to transmit digital signal first analog signal is converted into digital form known as Analog to digital conversion or (A/D Conversion)

A/D Conversion generally involves three steps:

1. Sampling
2. Quantization
3. Encoding

Sampling:

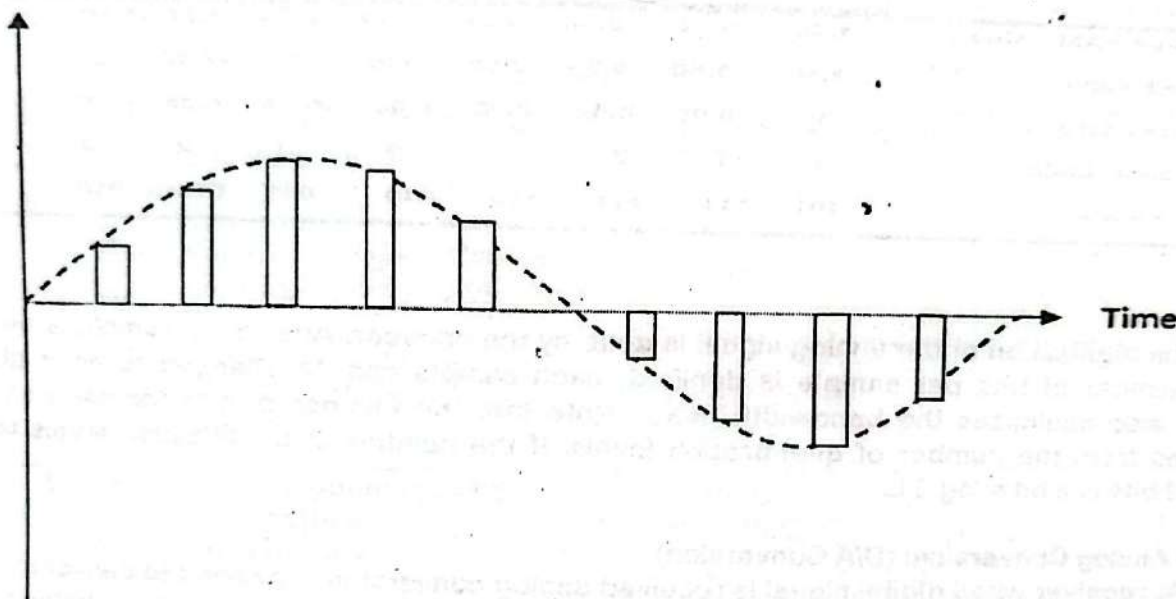
The first step in PCM is sampling. Sampling is a process of measuring the amplitude of a continuous-time signal at discrete instants, converting the continuous signal into a discrete signal. There are three sampling methods:

Ideal Sampling: In Ideal Sampling also known as instantaneous sampling pulses from the analog signal are sampled. This is an ideal sampling method and cannot be easily implemented.

Natural Sampling: Natural Sampling is a practical method of sampling in which pulse have finite width equal to Time period. The result is a sequence of samples that retain the shape of the analog signal.

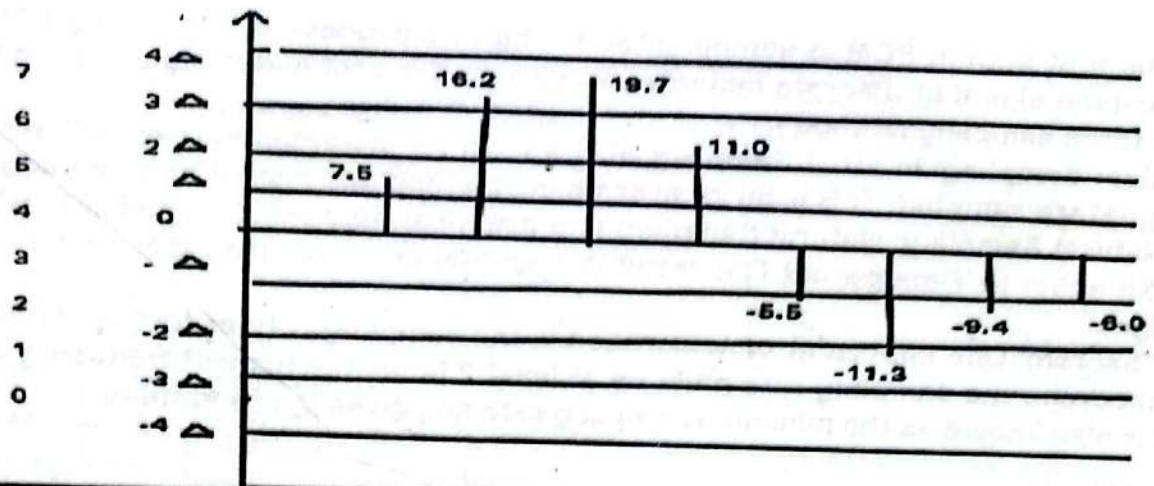
Theorem: One important consideration is the sampling rate or frequency. According to the Nyquist theorem, the sampling rate must be at least 2 times the highest frequency contained in the signal. It is also known as the minimum sampling rate and given by: $F_s = 2 \cdot f_{max}$

Amplitude

**Quantization:**

The result of sampling is a series of pulses with amplitude values between the maximum and minimum amplitudes of the signal. The set of amplitudes can be infinite with non-integral values between two limits. The following are the steps in Quantization:

- We assume that the signal has amplitudes between V_{max} and V_{min}
- We divide it into L or 2^N zones each of height d where, $d = (V_{max} - V_{min}) / L$
Where N are Number of output bits of N -Bit A/D Converter.
- The value at the top of each sample in the graph shows the actual amplitude.
- The normalized pulse amplitude modulation (PAM) value is calculated using the formula $\text{amplitude}/d$.
- After this we calculate the quantized value which the process selects from the middle of each zone.
- The Quantized error is given by the difference between quantized value and normalized PAM value.
- The Quantization code for each sample based on quantization levels at the left of the graph.



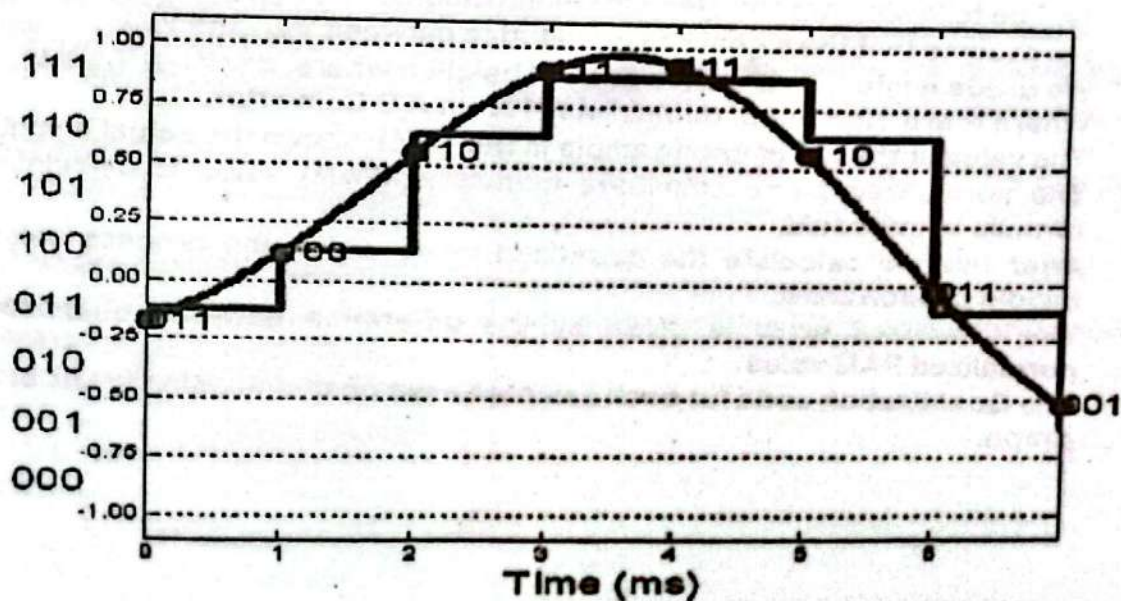
Normalised PAM value :	1.50	3.24	3.94	2.20	-1.10	2.26	-1.88	-1.20
Quantized Value :	1.50	3.50	3.50	2.50	-1.50	-2.50	-1.50	-1.50
Normalized Error :	0	0.26	-0.44	0.30	-0.40	-0.24	0.38	-0.30
Quantization Code :	5	7	7	6	2	1	2	2
Encoded Words :	101	111	111	110	010	001	010	010

Encoding

The digitization of the analog signal is done by the encoder. After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an n bit code. Encoding also minimizes the bandwidth used. Note that the number of bits for each sample is determined from the number of quantization levels. If the number of quantization levels is L , the number of bits is $n \text{ bit} = \log_2 L$.

Digital to Analog Conversion: (D/A Conversion)

At receiver when digital signal is received analog conversion is needed to deliver message effectively. In order to do Digital to Analog Conversion. This conversion involves translating the N -bit digital values into corresponding quantization levels, and maintaining this voltage for one sample duration, resulting in a signal resembling a series of stair steps.



The thick black line represents the reconstructed analog signal, accompanied by the 3-bit digital word that represents each sample. The continuous line depicts the original analog signal. Even with the application of filtering to smoothen the reconstructed signal and eliminate its stair-step appearance, it still deviates from the original signal, resulting in what is known as quantization error.

To reduce quantization error, one can enhance the accuracy by increasing the number of bits, denoted as N , for each sample. This adjustment reduces the size of the quantization intervals, leading to a more precise representation.

Chapter 14

Multiple Choice Questions

MCQ'S		ANS.
1. In Radio and Television broadcast, the information signal is in the form of: a) analog signal b) digital signal c) Both analog & digital signals d) neither analog nor digital signal		a
2. A communication channel consists of a) transmission line only b) optical fiber only c) free space only d) All of them		d
3. Voltage signal generated by a microphone is: a) digital in nature b) analog in nature c) hybrid in nature d) consists of bits & bytes		b
4. As compared to sound waves frequency of radio waves is: a) higher b) equal c) lower d) may be higher or lower		a
5. The process of superimposing signal frequency on the carrier wave is known as: a) transmission b) detection c) reception d) modulation		d
6. What is the frequency range of signals that can be transmitted in case of twisted pair of wires? a) 10MHz. to 15MHz. c) 100Hz. to 5MHz. b) 5MHz. to 10MHz. d) 10Hz. to 100Hz		c
7. The maintenance of a satellite's orbital position is called: a) maintenance keeping b) station keeping c) station maintenance d) attitude control		b
8. Process of mapping the sampled analog voltage values to discrete voltage levels is called: a) sampling c) quantizing b) sampling frequency d) encoding		c

9. AM is used for broadcasting because: a) It requires less transmitting power compare with other systems b) it is more noise immune than other modulation system c) No other modulation can provide the necessary bandwidth faithful transmission d) its use avoids receiver complexity	c
10. Data in compact disc is stored in the form of: a) analog signal b) digital signal c) noise d) both (a) &(b)	b

Carrier Wave	$V_c(t) = A \sin(2\pi f_c t + \phi)$ OR $c(t) = A_c \cos(2\pi f_c t)$
Message or Modulating Wave	$V_m(t) = B \sin 2\pi f_m t$ OR $m(t) = A_m \cos(2\pi f_m t)$
Amplitude Modulation	$AM(t) = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$ $AM(t) = A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$
Frequency Modulation	$FM(t) = f_c + k A_m \cos(2\pi f_m t)$ $FM(t) = f_c + k m(t)$
Modulation Index	$\mu = \frac{A_m}{A_c}$